

Roll Number: _____

Name: Solutions

Quiz 2
EE 3203: Power Electronics

Total Marks: 20

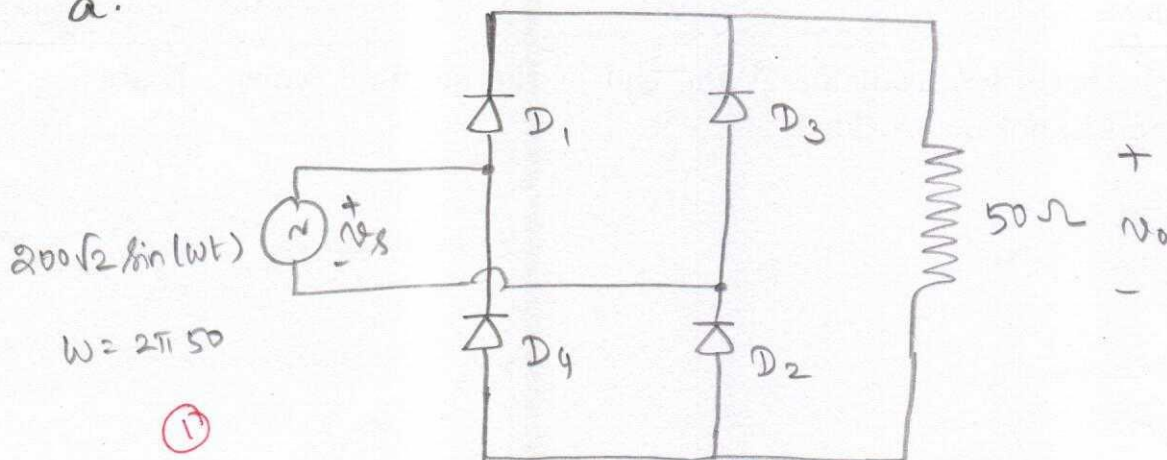
Time: 50 mins

Instructions

1. Write your roll number and name at the top of this question paper.
2. Check whether the question paper has all the pages.
3. The answers are to be filled in the space provided below the questions.
4. Use only pencil for sketching any waveforms/figures.
5. Graph sheets have been provided wherever required, to sketch the waveforms.
6. Rough sheets are available at the end of the question paper. Extra papers will not be provided.

1. A single phase full bridge diode rectifier feeds a resistive load of $50\ \Omega$ from a 200 V(rms) , 50 Hz sinusoidal input supply $v_s(\omega t)$. (4)
- Draw a neat sketch of the single phase full bridge diode rectifier with the purely resistive load of $50\ \Omega$ connected across the output terminals.
 - Sketch the input supply $v_s(\omega t)$ and the output voltage $v_o(\omega t)$ of the rectifier.
 - Indicate the conducting devices at various instants of the waveform $v_o(\omega t)$.
 - Compute the power drawn by the load.

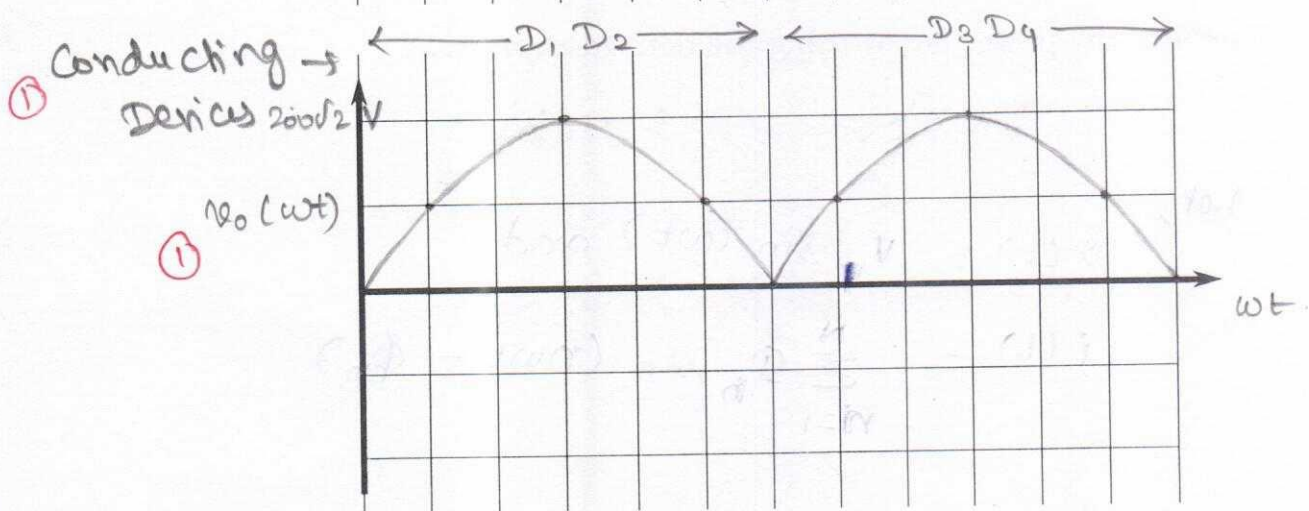
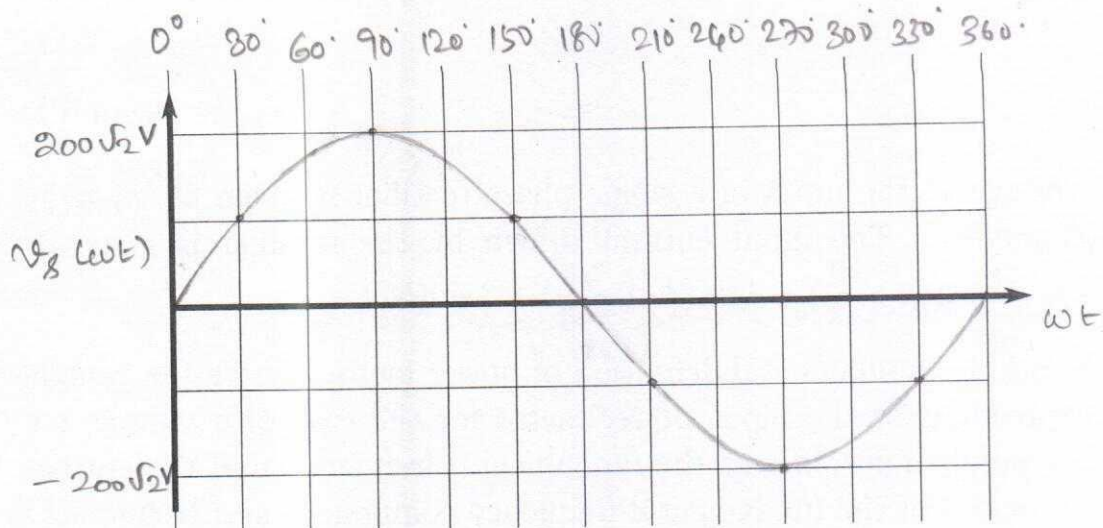
a.



d. Power drawn = $\frac{V_{rms}^2}{R} = \frac{200^2}{50} = 800\text{ W}$.

(17)

Power drawn = 800 W.



2. The voltage at the input of a single phase rectifier is given by, $v(\omega t) = 230\sqrt{2}\sin(\omega t)$. The input current drawn by the rectifier is given by $i(\omega t) = 10\sin(\omega t - \frac{\pi}{3}) + 4\sin(3\omega t - \frac{\pi}{6}) + 3\sin(5\omega t - \frac{\pi}{3})$. (4)
- Using the fundamental definition of power factor, derive the general expression for the input power factor for the case of a voltage being purely sinusoidal at the fundamental frequency and the current composed of the fundamental frequency component and harmonics.
 - Compute the power factor using the derived expression.

a. Let, $v(t) = V_m \sin(\omega t)$ and

$$i(t) = \sum_{n=1}^N I_n \sin(n\omega t - \phi_n)$$

By definition,

$$\text{Power factor} = \frac{\text{Real power}}{\text{Apparent power}}$$

$$= \frac{\left(\frac{V_1}{\sqrt{2}}\right)\left(\frac{I_1}{\sqrt{2}}\right)\cos(\phi_1)}{V_{\text{rms}} \cdot I_{\text{rms}}}$$

$$\Rightarrow \text{Power factor} = \frac{(V_1 I_1 \cos \phi_1 / 2)}{\left(\frac{V_1}{\sqrt{2}}\right) \sqrt{\sum_{n=1}^N \left(\frac{I_n}{\sqrt{2}}\right)^2}}$$

$$\Rightarrow \text{Power factor} = \frac{\left(\frac{P_1}{\sqrt{2}}\right) \cos \phi_1}{\sqrt{\sum_{n=1}^N \left(\frac{P_n}{\sqrt{2}}\right)^2}}$$

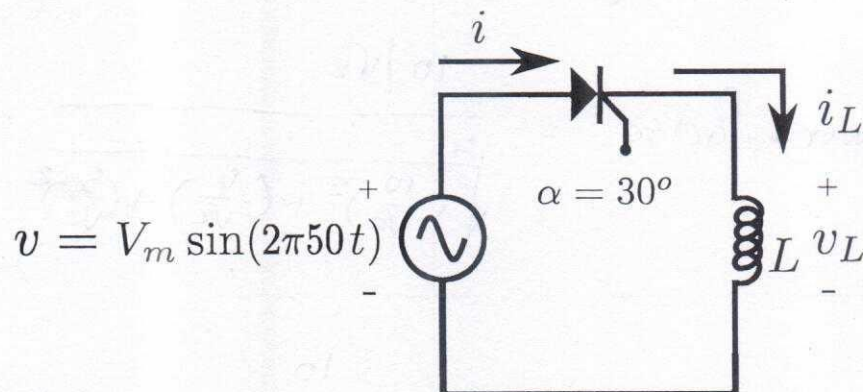
$$\Rightarrow \text{Power factor} = \frac{P_{1\text{rms}}}{P_{\text{rms}}} \times \cos \phi_1.$$

$$b. \quad \text{Power factor} = \frac{10/\sqrt{2}}{\sqrt{\left(\frac{10}{\sqrt{2}}\right)^2 + \left(\frac{4}{\sqrt{2}}\right)^2 + \left(\frac{3}{\sqrt{2}}\right)^2}} \cdot \cos\left(\frac{\pi}{3}\right).$$

$$= \frac{10}{\sqrt{100 + 16 + 9}} \times \frac{1}{2}.$$

$$\Rightarrow \text{Power factor} = 0.45$$

3. The figure shows a half wave controlled rectifier feeding a purely inductive load. Assuming a SCR firing angle of $\alpha = 30^\circ$. (4)
- Sketch the input voltage $v(\omega t)$ and the inductor voltage $v_L(\omega t)$.
 - Using the waveform, compute the extinction angle β (the angle at which the SCR ceases to conduct, the angle being measured from $\omega t = 0^\circ$).
 - Compute the circuit turn-off time t_c of the SCR.



(c) $\omega t_c = \pi/6$

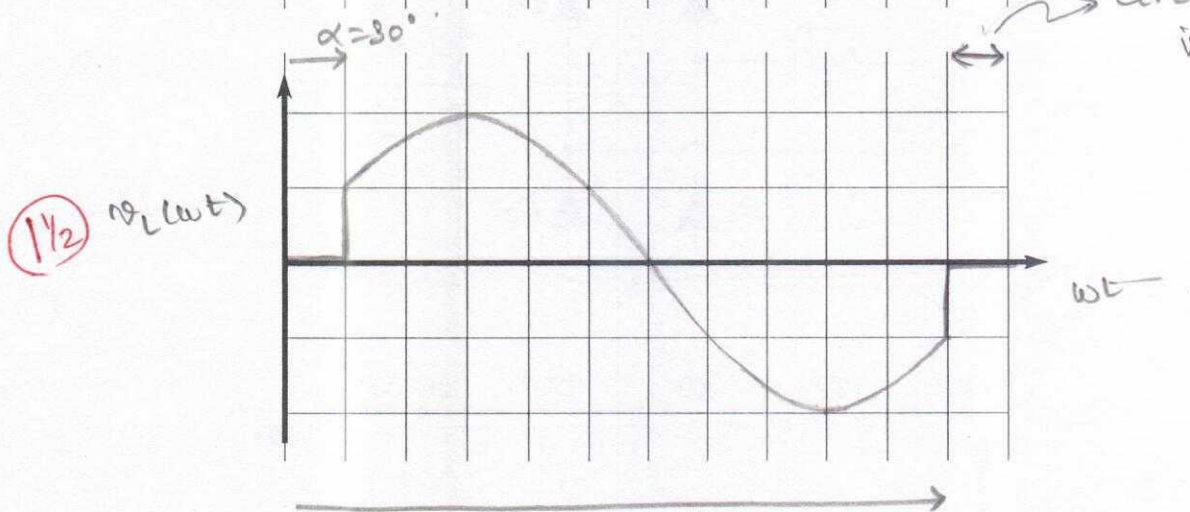
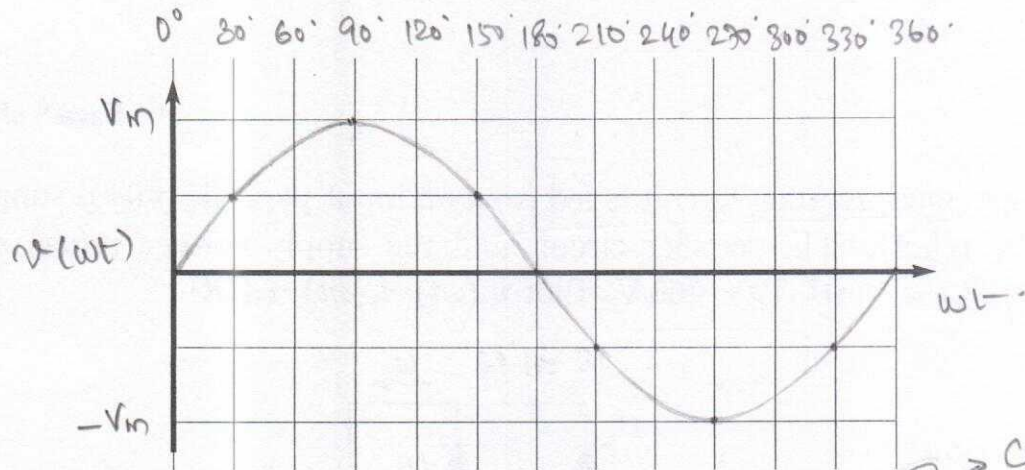
$$\Rightarrow t_c = \frac{\pi/6}{2\pi 50}$$

$$\Rightarrow t_c = \frac{1}{600}$$

(142)

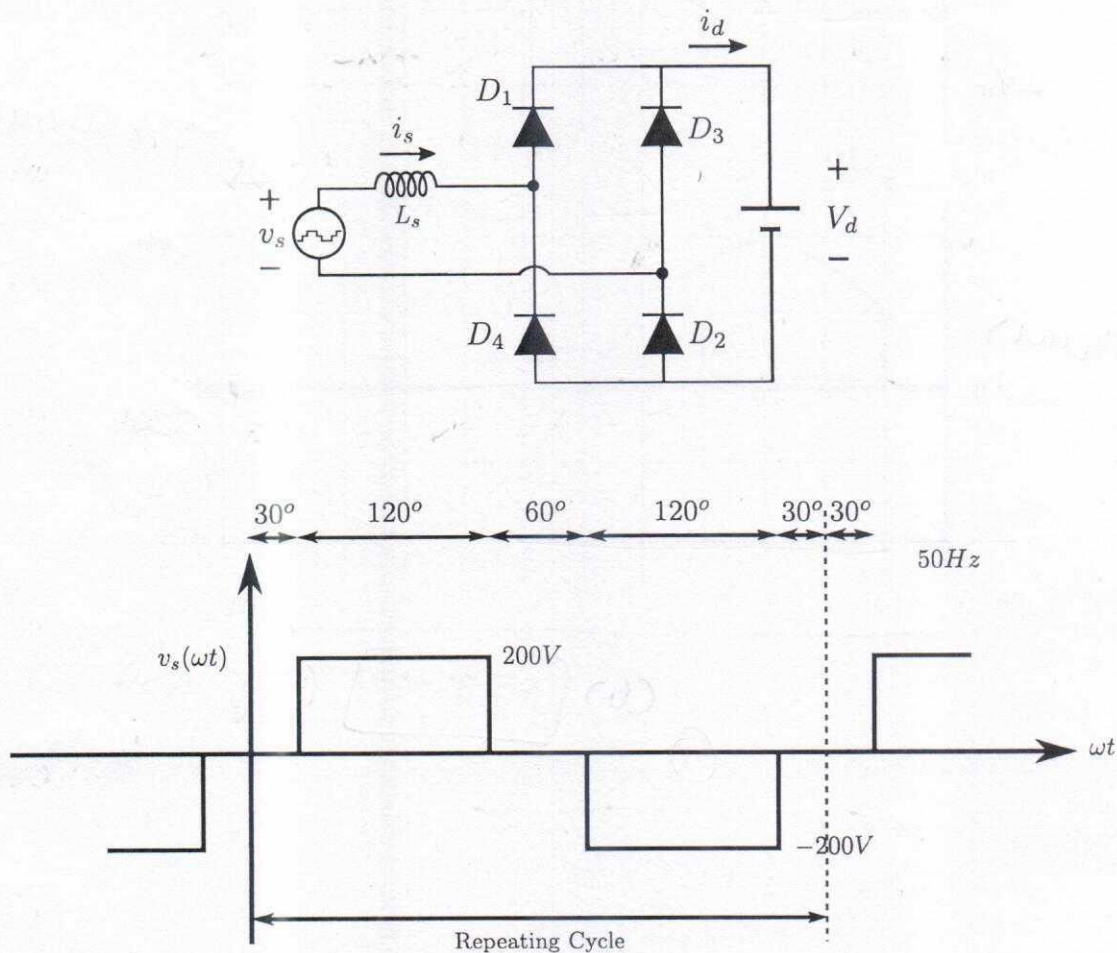
$$\Rightarrow t_c = 1.67 \text{ ms}$$

(a)

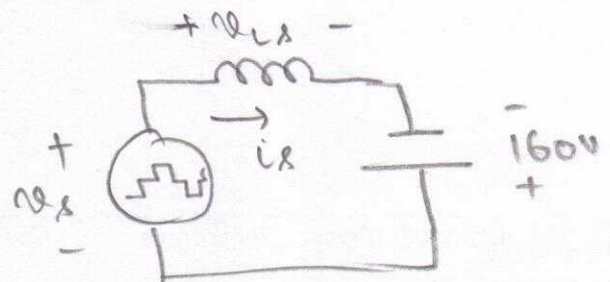
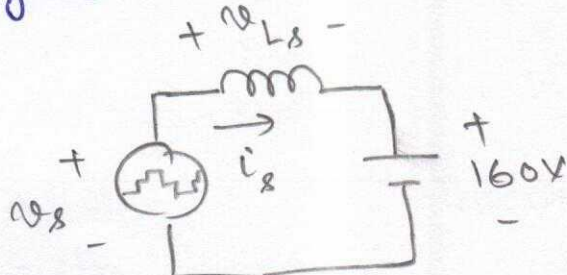


(b) $\beta = 330^\circ$ (By volt-sec balance)

4. A single phase rectifier circuit is fed from an input periodic pulsed supply voltage $v_s(\omega t)$. The rectifier circuit and the supply voltage $v_s(\omega t)$ are shown. $L_s = 1 \text{ mH}$, $V_d = 160 \text{ V}$. Plot $v_s(\omega t)$, $i_s(\omega t)$, $i_d(\omega t)$. (4)

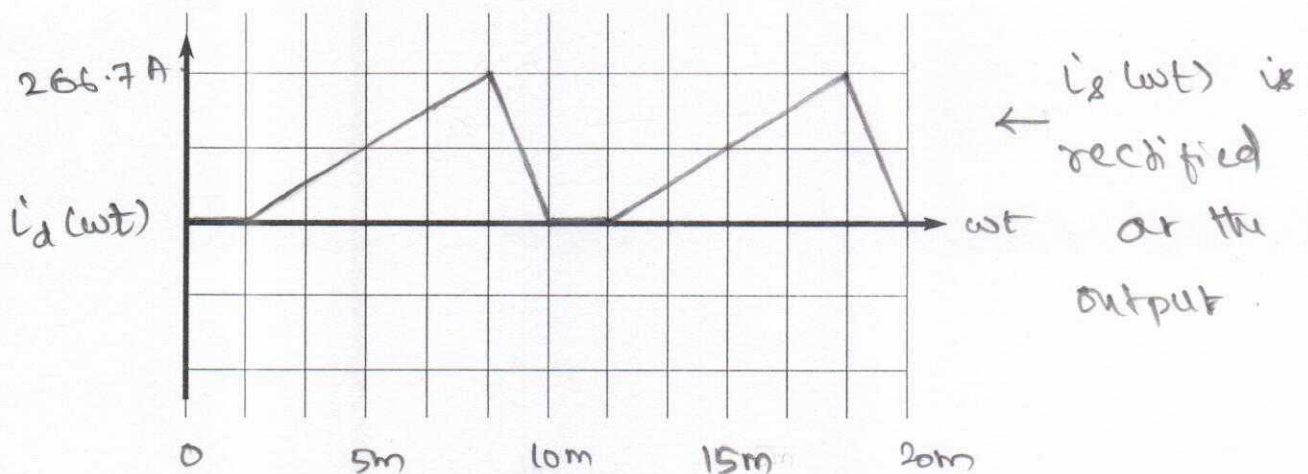
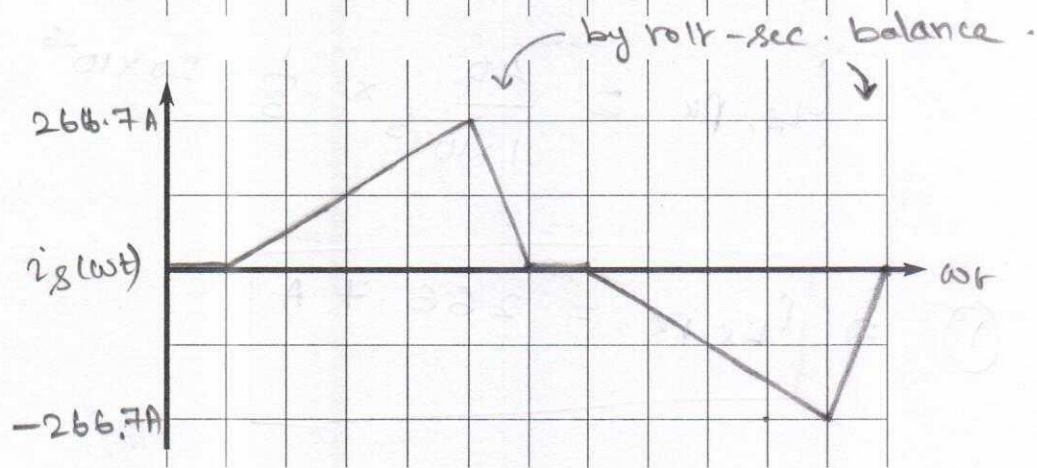
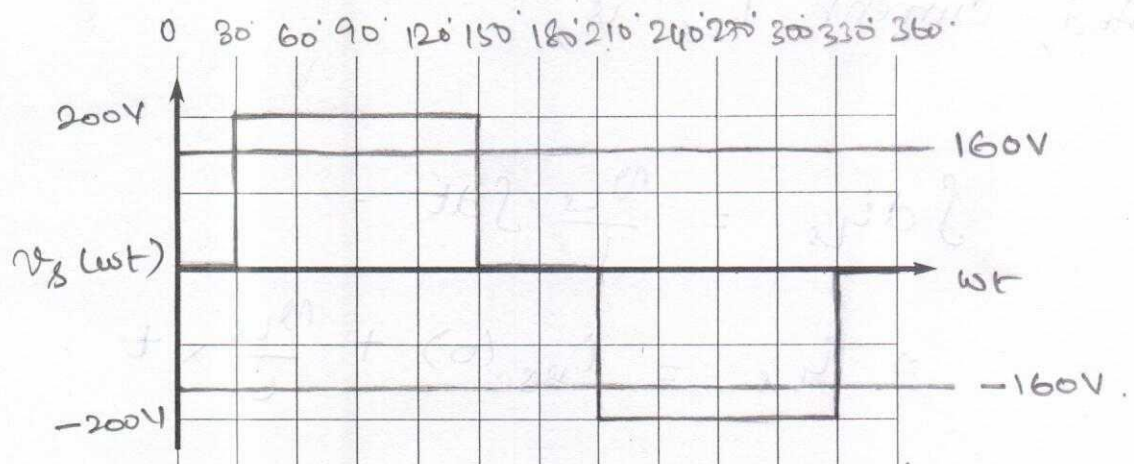


The equivalent circuits during the $+$ ve and $-$ ve half cycles are,



$$0 \leq \omega t \leq 180^\circ$$

$$180^\circ \leq \omega t \leq 360^\circ$$



So, from $30^\circ \leq \omega t \leq 150^\circ$, $v_{Ls} = 40V$.

$150^\circ \leq \omega t \leq 180^\circ$, $v_{Ls} = -160V$.

$210^\circ \leq \omega t \leq 330^\circ$; $v_{Ls} = -40V$.

$330^\circ \leq \omega t \leq 360^\circ$; $v_{Ls} = +160V$.

The $\frac{di_{Ls}}{dt}$'s are given by $\frac{v_{Ls}}{L}$
 $L = 1mH$.

And, current i_{Ls} is,

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$$\int di_{Ls} = \frac{V_{Ls}}{L} \int dt$$

$$\Rightarrow i_{Ls} = i_{Ls}(0) + \frac{V_{Ls}}{L} \times t$$

$$i_{Ls, pk} = \frac{40}{1 \times 10^{-3}} \times \frac{1}{3} \times 20 \times 10^{-3} = \frac{800}{3} \text{ A}$$

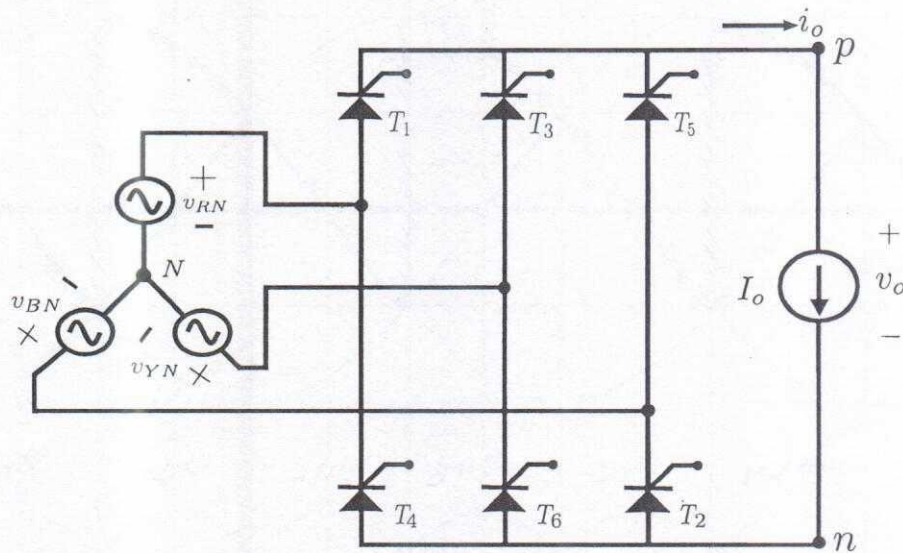
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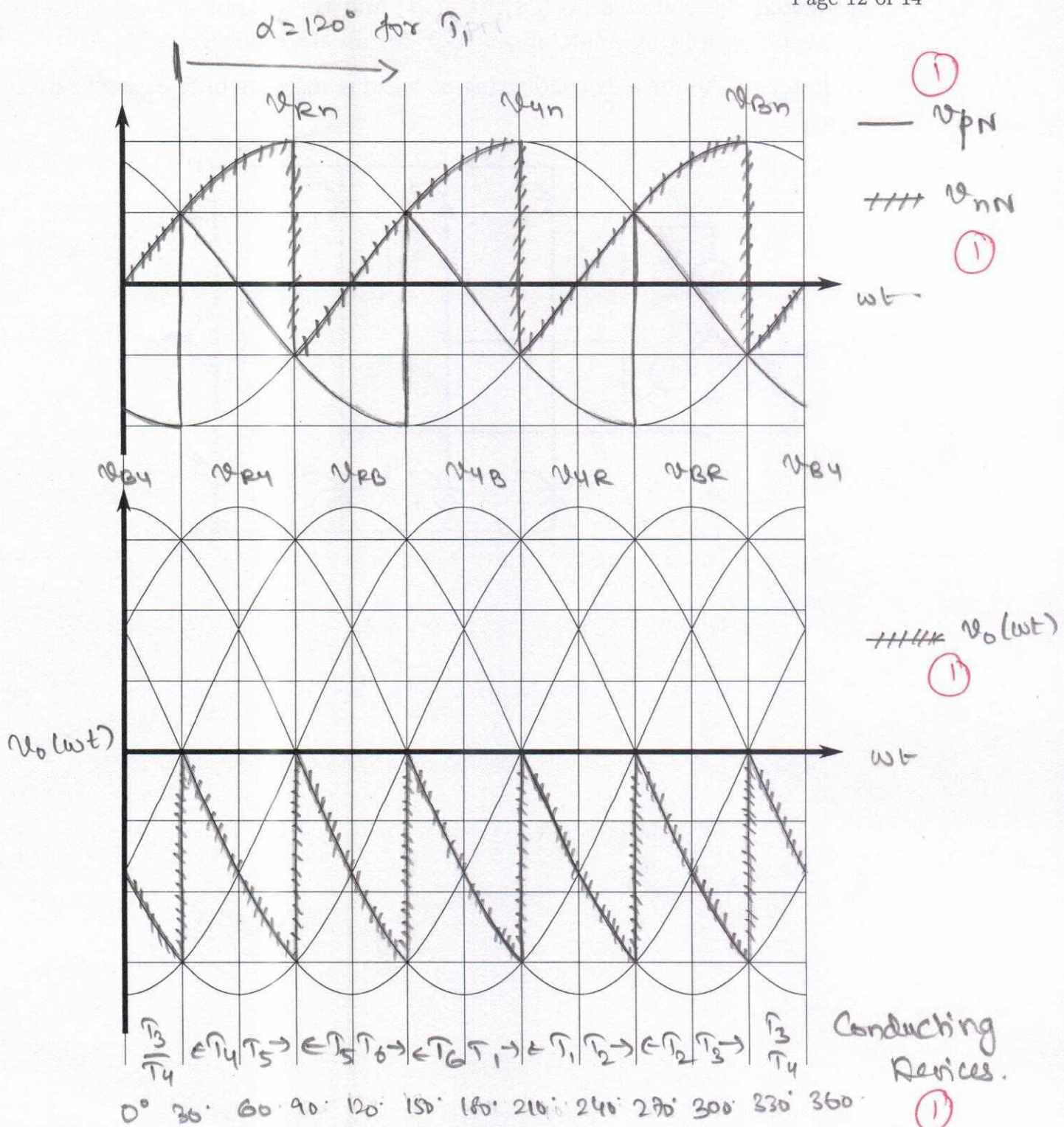
$$\Rightarrow i_{Ls, pk} = 266.7 \text{ A}$$

5. A three phase thyristor bridge rectifier is shown below.

(4)

- Sketch the voltages $v_{pN}(\omega t)$, $v_{nN}(\omega t)$ and the output voltage $v_o(\omega t)$ assuming a firing angle of $\alpha = 120^\circ$ for all the devices.
- Indicate the devices conducting at various instants of the waveform.





Rough Work

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